

Sang-Hyeon Park

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Current affiliation: Agency for Defense Development, South Korea

Education

Hanyang University, Seoul, South Korea

Mar. 2020 – Feb. 2024

B.S. in Energy Engineering

GPA: 4.36/4.5 (3.92/4.0, 162 credits), *Summa Cum Laude* (Ranked 1st in Class)

Research Interest

in-silico materials science / Energy materials / Electronic inorganic materials / XAI for materials /

Solid-state synthesis / Interfacial thermodynamics / Diffusion and kinetics / Defect chemistry / Multiscale modeling

Research Experience

Agency for Defense Development (ADD), Daejeon, South Korea

May 2024 – Present

Research Officer for National Defense (ROND), ROK Army

- Research in materials chemistry and informatics
 - Structure–property relationship in energy storage materials through *in-silico* computation
- Materials informatics through big data construction and interpretable machine learning
 - Development in energy storage/conversion devices for military applications
- Designing electrode and electrolyte materials for reserve thermal batteries
- Membrane–electrode assembly for PEMFC

Korea Institute of Science and Technology (KIST), Seoul, South Korea

Sep. 2023 – Feb. 2024

Full-time Undergraduate Student Intern (Advisor: Dr. Seungho Yu)

- Trained in computational approaches for analysis and design of sulfide solid electrolytes

Publications

7 Lead-author / 4 Co-author; 4 Published / 4 Under Review / 3 In Preparation

† equal contribution, * corresponding author

1. **Finite Element Investigation of Thermal Stability in Low Melting Electrolytes for Thermal Batteries**
SH Park, H Won, JH Cho, CN Im, HK Yoon and JS Yeo*
Manuscript In preparation (planned submission: Aug. 2026)
2. **Collision Integrity Evaluation of Stacked Thermal Battery Cells: Pellet Compression Behavior and FEM Analysis**
 J Kim†, SH Park†, H Roh, S Lee, HS Kim and TY Ahn*
Manuscript In preparation (planned submission: Aug. 2026)
3. **Theoretically investigated phase evolution of AFeF₃ (A=Li, Na) polymorphs demonstrated by simple mechanochemical synthesis**
SH Park†* and H Won†
Manuscript In preparation (planned submission: Sep. 2026)
4. **Transferable descriptor-guided interfacial thermodynamics and electrochemical stability for stable solid-state alkali battery interfaces**
SH Park, H Won, S Han, HK Yoon and J Lee*
Submitted, Energy Storage Materials
5. **Reaction-Controlled Fluorination of FeS₂ Cathodes via FeF₃ Surface Layer Formation for High-Voltage Thermal Batteries**
 HR Yu, IW Lee, TY Ahn, WJ Ahn, SH Park, CN Im, HK Yoon, Y Choi and YS Lee*
Under review, Applied Surface Science
6. **Approaching Intrinsic High-Temperature Performance in Garnet-Type Solid Electrolyte (LLZTO): Microstructure and Surface-Chemistry Engineering for Advanced Salt-free Thermal Batteries**
 JH Lim, SH Park, C Yoon, TY Ahn, H Roh, HK Yoon and J Lee*
Under review, Journal of Power Sources

7. **Stabilizing Interfaces in Magnesium Solid-State Batteries via High-throughput Coating Design**
SH Park^{*}, C Youn, JH Lim, G Kim, HK Yoon, Y Choi and J Lee^{*}
Under review, Small Methods
8. **Data-Driven Design Strategies for Chemically Stabilizing Cathode–Coating Interfaces Employing Interpretable Machine Learning**
SH Park, D Kang, HK Yoon, Y Choi and J Lee^{*}
Journal of Materials Chemistry A, 14 (39), 26011-26025, (2026).
 [Featured on the Back Cover]
9. **Impact reliability of multilayer stacked-cell thermal batteries using 3D FEM simulation and gas gun test validation**
 J Kim[†], H Kang[†], TY Ahn, **SH Park**, M Kim, SH Kang, H Roh, S Lee and HS Kim^{*}
Journal of Materials Research and Technology, 42, 5060-5072, (2026).
10. **First-Principles Insights into the Role of 3d Transition-Metal Substituents in Intercalating Li_xFeF_3 Cathode Materials for Lithium-Ion Batteries**
SH Park, JH Lim, TY Ahn, HK Yoon, J Lee^{*} and H Roh^{*}
ACS Applied Energy Materials, 8 (19), 14712-14724, (2025).
11. **Comparative study of Ta and Ga doping effects on $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ garnet electrolytes for advanced thermal battery applications**
 H Kang, H Roh, J Lee, **SH Park**, J Park, H Jeong, HK Yoon, TY Ahn^{*} and Y Choi^{*}
RSC advances, 15 (33), 27345-27355, (2025).

Conference Presentations

1. **Data-driven Evaluation of interfacial stability in Batteries: From First-Principles to Military Applications**
 - Oral presentation in Korean Institute of Military Science and Technology Spring meeting (2026)
2. **First-principles investigated transition metal substitution into Li_xFeF_3**
 - Poster presentation in Korean Ceramic Society Spring meeting (2025)

Honor, Awards, and Scholarships

Honors

- **Research Officer for National Defense (ROND)** May 2024 – Present
 - First Lieutenant in the Republic of Korea Army
 - Selected as one of 20 research officers nationwide for STEM-based national defense research
- **Valedictorian** Feb. 2024
 - Ranked 1st in Class
 - Distinguished Graduate Award for outstanding extracurricular contributions
- **Academic Excellence Award (Top 1% Academic Achievement)** Jun. 2023
 - Hanyang University, College of Engineering

Awards

- **2nd Prize, 2023 MiliTECH Challenge** Dec. 2023
 - Hosted by KAIST; Awarded by the Agency for Defense Development
- **1st Prize, National Undergraduate Chemical Engineering Design Competition** Nov. 2023
 - Hosted by the Korean Institute of Chemical Engineers
 - Awarded by SK geocentric Inc.
- **2nd Prize, Energy and Environment Exploration Competition** Nov. 2022
 - Hosted and awarded by the SAMSUNG engineering Inc.

Scholarships

- **National Science and Technology Scholarship** Mar. 2022 – Feb. 2024

- Korea Student Aid Foundation (KOSAF)

- **Readership Scholarship** Dec. 2023
- Hanyang University Library, The Champion of 2023 Hanyang University Reading Quiz Competition
- **Leadership Scholarship** Mar. 2020 – Feb. 2022
- Hanyang University, Student President in Department of Energy Engineering
- **Academic Excellence Scholarship** Aug. 2020
- Hanyang University

Skills

Experimental – Electrochemical spectroscopy impedance (EIS), mechanochemical synthesis, coin-cell fabrication, atomic layer deposition, and electrochemical testing/cycling

Computational – Quantum Espresso, VASP, LAMMPS, COMSOL Multiphysics, machine learning, and python

Extracurricular Activities

TBD

References

TBD